Jungeui Choi

São Paulo, Brazil | +55 (11) 99902-8788 | jungchoi0221@gmail.com | linkedin.com/in/jungeuichoi

AI Engineer and Ph.D. researcher specializing in deep learning, medical imaging, and production LLM systems, with peer-reviewed research on inverse problems and 3D image segmentation.

Work Experience

AI Engineer — *Jul 2025 – Present*
Scopic Software — Remote (US)

- Designed and built a multi-tenant AI agent platform with LangGraph agent workflows, retrieval-augmented generation (RAG) over Qdrant, and real-time SSE streaming
- Engineered a FastAPI backend with schema-isolated PostgreSQL multi-tenancy, Celery task queues, and Redis caching
- Built an automated 3D medical image segmentation pipeline (CT → bone segmentation → 3D surface models) using nnU-Net, with DVC and MLflow integration for reproducible MLOps

Software Engineer — *Jan 2025 – Jun 2025*
Dtlabs — Bauru, Brazil

- Developed visual odometry, SLAM, and ML-based computer vision algorithms for aerial image processing
- Implemented 3D terrain reconstruction, image stitching, and georeferencing optimization for aerial mapping

CFD Engineer — *Feb 2024 – Dec 2024*
Numerical Offshore Tank (TPN) – University of São Paulo — São Paulo, Brazil

- Performed computational fluid dynamics simulations of ship hydrodynamics using OpenFOAM and Star-CCM+

Education

Escola Politécnica, Universidade de São Paulo — *Jan 2025 – Present*
Ph.D. in Mechanical Engineering (Mechatronics) — São Paulo, Brazil

Escola Politécnica, Universidade de São Paulo — *Jan 2020 – Dec 2024*
B.Sc. in Naval Architecture and Marine Engineering — São Paulo, Brazil

Projects

Deep Learning for the 3D EIT Inverse Problem (Ph.D. Research) — *Jan 2025 – Present*
Mechatronics Engineering Department, University of São Paulo — São Paulo, Brazil

- Developing 3D image reconstruction methods for Electrical Impedance Tomography using implicit neural representations, diffusion models, normalizing flows, KANs, and transformers
- Built the full research pipeline: CT segmentation, 3D FEM meshing, forward simulation, and neural inverse solver

Automatic Lung Segmentation via ROIFT — *Mar 2021 – Mar 2022*
Mechatronics Engineering Department, University of São Paulo — São Paulo, Brazil

- Developed automatic seed generation and 3D segmentation of chest CT images using the Relaxed Oriented Image Foresting Transform (ROIFT)

AI Stock Trading Bot — *Sep 2023 – Nov 2023*
Huawei Cloud Hackathon — São Paulo, Brazil

- Built an NLP pipeline for real-time news sentiment analysis driving automated trading; **Winner**, LAC ICT Talent (Mexico)

Publications

- J. Choi *et al.*, "On the Implicit Representation of the Electrical Impedance Tomography Inverse Problem," *IEEE*, 2026
- J. Choi *et al.*, "Sinusoidal Representation of the Electrical Impedance Tomography Inverse Problem," *IEEE*, 2026
- J. Choi *et al.*, "Lung Automatic Seeding and Segmentation: A Robust Method Based on Relaxed Oriented Image Foresting Transform," *Biomedical Signal Processing and Control*, vol. 117, 2026
- J. Choi *et al.*, "Automatic Lung Segmentation with Seed Generation and ROIFT Algorithm for the Creation of Anatomical Atlas," *ICGG*, Springer, 2022

Skills

Languages: Python, C++
AI & Machine Learning: PyTorch, JAX, LangGraph, RAG (Qdrant), nnU-Net, MLflow, DVC
Backend & Infrastructure: FastAPI, PostgreSQL, Redis, Celery, Docker, AWS
Web & Simulation: Next.js, React, OpenFOAM, Star-CCM+